# Comprehensive Guide to Chronic Myeloid Leukemia (CML)

## 1. Overview and Definition

Chronic myeloid leukemia (CML), also known as chronic myelogenous leukemia, is a type of cancer that originates in the blood-forming pluripotent stem cells of the bone marrow and subsequently invades the blood [cite: 1.1.1, 1.1.3, 1.1.4]. The disease is characterized by malignant transformation and clonal myeloproliferation, resulting in a dramatic overproduction of both mature and immature granulocytes (a type of white blood cell) [cite: 1.1.1]. The "chronic" designation indicates that this cancer typically worsens slowly without treatment, compared to acute leukemias [cite: 1.1.3, 1.2.1].

### Epidemiology and Statistics

- In the United States, there were approximately 9,560 new cases of CML and 1,290 deaths estimated in 2025 [cite: 1.1.1].
- The average age of a patient diagnosed with CML is 66 years [cite: 1.1.1].
- The average lifetime risk of developing CML in the US among both sexes is approximately 0.2%, or 1 in 500 Americans [cite: 1.1.1].

## 2. Pathophysiology and Genetics

CML is almost exclusively driven by a specific, acquired genetic abnormality. It is not an inherited condition passed down from birth parents to children; rather, it develops over time [cite: 1.2.1].

- **The Philadelphia (Ph) Chromosome:** Present in 90% to 95% of CML cases, this is the hallmark genetic marker of the disease [cite: 1.1.1]. It is formed by a reciprocal translocation between chromosome 9 and chromosome 22, denoted as t(9;22) [cite: 1.1.1, 1.2.5]. A piece of chromosome 9 breaks off and trades places with a piece of chromosome 22, resulting in a shortened chromosome 22 known as the Philadelphia chromosome [cite: 1.2.1].

- **The BCR::ABL1 Fusion Gene:** The translocation fuses the ABL oncogene from chromosome 9 with the BCR gene on chromosome 22 [cite: 1.1.1].
- **Oncoprotein Production:** The resulting BCR::ABL1 fusion gene instructs the cell to produce a mutated oncoprotein called BCR::ABL1 tyrosine kinase [cite: 1.1.1]. This protein leads to uncontrolled cellular proliferation and continuous growth of abnormal white blood cells (leukemic cells), which crowd out healthy, normal cells [cite: 1.1.1, 1.1.3, 1.2.1].

## 3. Signs and Symptoms

Because CML is a slow-progressing disease, many patients are completely asymptomatic during the early stages, and the diagnosis is often discovered incidentally during routine blood work [cite: 1.1.1, 1.1.3]. When symptoms do appear, they develop insidiously over months to years.

- **Early/Chronic Phase Symptoms:** Malaise, fatigue, weakness, anorexia, unexplained weight loss, drenching night sweats, and gouty arthritis [cite: 1.1.1, 1.1.2, 1.1.5]. Patients may also experience a sense of abdominal fullness, particularly in the left upper quadrant, due to an enlarged spleen (splenomegaly) [cite: 1.1.2]. Leukostasis can occasionally cause tinnitus, stupor, and urticaria [cite: 1.1.2].
- **Advanced Disease Symptoms:** As the disease progresses to accelerated or blast phases, ominous signs emerge. These include progressive splenomegaly (seen in 60-70% of cases), hepatomegaly, pallor, easy bruising and bleeding (due to thrombocytopenia), fever, lymphadenopathy, bone pain, and the development of destructive bone lesions [cite: 1.1.1, 1.1.2, 1.1.5].

## 4. Diagnostic Procedures

A comprehensive workup is essential for diagnosing CML and determining the phase of the disease.

- **Blood Tests:** A complete blood count (CBC) with differential to assess red blood cells, white blood cells, and platelets, alongside blood chemistry studies [cite: 1.1.3, 1.1.5].
- **Bone Marrow Aspiration and Biopsy:** Useful for evaluating the percentage of blasts and checking for signs of accelerated phase or blast crisis, myelofibrosis, ringed sideroblasts, or red cell aplasia [cite: 1.1.2, 1.1.5]. The necessity of bone marrow testing for all newly diagnosed, routine chronic-phase patients is questionable outside of clinical trials, but it remains appropriate for those with clinical signs of advanced disease [cite: 1.1.5].
- **Cytogenetic Analysis:** Performed on bone marrow or blood to confirm the presence of the Philadelphia chromosome and detect any additional chromosomal abnormalities that might impact treatment response [cite: 1.1.3, 1.1.5].
- **Fluorescence in situ Hybridization (FISH):** Can be used on blood or marrow to confirm the BCR::ABL1 translocation, even when classic cytogenetic abnormalities are absent [cite: 1.1.2, 1.1.5].
- **Reverse Transcription-Polymerase Chain Reaction (RT-PCR):** The most sensitive diagnostic technique available. It detects the BCR::ABL1 rearrangement at a molecular level [cite: 1.1.2, 1.1.5]. Patients who test negative for BCR::ABL1 via RT-PCR have a disease more consistent with chronic myelomonocytic leukemia [cite: 1.1.5].

## 5. Phases of CML

CML generally progresses through three distinct clinical phases. The majority of patients in developed countries are diagnosed in the early, chronic phase [cite: 1.1.3, 1.1.4, 1.2.5].

- **Chronic Phase:** An initial indolent period that may last 5 to 6 years if untreated [cite: 1.1.2]. Most patients are diagnosed here and, if properly treated, can remain in this phase without progressing [cite: 1.1.4].

- **Accelerated Phase:** Indicated by treatment failure, worsening anemia, progressive thrombocytopenia or thrombocytosis, persistent or worsening splenomegaly, clonal evolution, increasing blood basophils, and an increase in bone marrow or blood blasts up to 19% [cite: 1.1.2, 1.1.5].
- **Blast Phase (Blast Crisis):** An abrupt or gradual progression where blasts reach 20% or more in the peripheral blood or bone marrow [cite: 1.1.5]. The disease behaves aggressively, akin to acute leukemia, presenting with severe symptoms like extreme thrombocytopenia, increasing pain, fever, and bone lesions [cite: 1.1.5, 1.2.1].

## 6. Treatment Options

The landscape of CML treatment has been revolutionized by targeted therapies, transitioning the disease from requiring intensive chemotherapy and bone marrow transplants to being managed primarily with oral medications [cite: 1.1.3]. The primary goal of therapy is to reduce the number of CML cells with the BCR::ABL1 gene to 1% or less, ideally as close to zero as possible, to prevent progression to the accelerated or blast phases [cite: 1.2.1, 1.2.5].

### Tyrosine Kinase Inhibitors (TKIs)

TKIs are the optimal front-line treatment for chronic-phase CML [cite: 1.1.5]. They specifically target and stop the activity of the BCR::ABL1 protein [cite: 1.2.1]. While rarely curative alone, they dramatically improve response rates and prolong survival [cite: 1.1.1, 1.1.2].

- **First-generation:** Imatinib mesylate [cite: 1.1.3, 1.1.5].
- **Second and Third-generation:** Nilotinib, Dasatinib, Bosutinib, and Ponatinib offer greater potency and selectivity for BCR::ABL1 [cite: 1.1.2, 1.1.5].
- **Asciminib:** Indicated for newly diagnosed CML in the chronic phase, specifically for patients with certain mutations [cite: 1.1.2].

*Note:* Bariatric surgery can impede the proper absorption of oral TKIs, potentially leading to suboptimal therapeutic responses [cite: 1.1.5].

### Other Treatment Modalities

- **Allogeneic Stem Cell Transplantation (SCT):** Reserved for patients with accelerated or blast-phase CML who are resistant to BCR::ABL1 inhibitors [cite: 1.1.2]. While it offers a potential cure, it comes with substantial morbidity and a 5% to 10% treatment-related mortality rate. Long-term survival post-SCT is comparable to TKI drug treatment [cite: 1.1.2, 1.1.5].
- **Interferon Alfa:** Historically used and still a viable option. Approximately 10% to 20% of patients achieve a complete cytogenetic response, though maintenance therapy is required and side effects can be limiting [cite: 1.1.5, 1.2.4].
- **Hydroxyurea and Chemotherapy:** Used as supportive or palliative agents in advanced disease stages [cite: 1.1.1, 1.2.4].

## 7. Treatment Monitoring and Response Criteria

Monitoring the response to TKI therapy is the most critical prognostic factor for CML patients. Discontinuation of TKIs is feasible in selected patients with careful monitoring, but TKI withdrawal syndrome (characterized by musculoskeletal pain) affects about 25% of patients who stop therapy [cite: 1.1.2, 1.2.2, 1.2.5].

### Monitoring Schedule and Categories

- Measurements are taken at baseline, 3 months, 6 months, and 1 year [cite: 1.1.2]. The NCCN guidelines suggest PCR testing monthly for the first 6 months, then bimonthly for months 7 through 12, and quarterly thereafter [cite: 1.2.2].
- Treatment response is categorized. The ELN uses "optimal," "warning," and "failure," while the NCCN categorizes responses as "TKI-sensitive disease," "possible TKI resistance," and "TKI-resistant" [cite: 1.2.2].

### Milestones of Response

- **Cytogenetic Response:** Measures the reduction of Ph+ chromosome cells [cite: 1.1.2].
- **Molecular Response:** Measured by quantitative PCR (qPCR) detecting BCR::ABL1 protein levels. A **Major Molecular**

**Response (MMR)** is achieved when blood BCR::ABL1 levels fall to less than 1/1000th (≤0.1%) of the expected value for untreated CML [cite: 1.1.2, 1.1.3]. At this level, there is only one Ph+ CML cell per 1 million normal cells [cite: 1.1.3].

- If MMR is lost, prompt resumption of TKI therapy is highly recommended [cite: 1.2.2].

## 8. Special Considerations: Pregnancy and CML

Managing CML during pregnancy presents a complex clinical challenge, balancing maternal treatment benefits against fetal risks. CML treatment is often lifelong, intersecting with many patients' family planning goals [cite: 1.2.4].

- **TKI Teratogenicity:** TKIs are known to have teratogenic effects and can cause fetal malformations. Discontinuation of TKI therapy is strongly recommended if the patient has achieved a Major Molecular Response (MMR) [cite: 1.2.4].
- **Timing of Treatment:** Previous studies advised strictly against TKI use before 15 weeks of gestation. However, newer research suggests that TKI use might be considered after organogenesis, contingent upon a meticulous risk-benefit analysis [cite: 1.2.4].
- **Contraindicated Drugs:** Dasatinib is explicitly not recommended during pregnancy due to the lack of sufficient safety data [cite: 1.2.4].
- **Alternative Options:** If treatment is deemed necessary to prevent CML progression during pregnancy, alternative treatments like Interferon (IFN) or hydroxyurea may be considered based on continuous monitoring for disease relapse and recurrence [cite: 1.2.4]. Pregnancy can proceed safely if the disease is in the chronic phase and progressing favorably toward an MMR [cite: 1.2.4].